Data Structures Lab (CL2001)

Date: October 15th 2026

Section: H

Sessional Exam

Total Time (Hrs):	**2 hrs**
Total Marks:	**25**
Total Questions:	**2**

Roll No Section Student Signature

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Attempt all the questions.

CLO #: 1

Q1: [10 Marks]

Given an array of integers, find the **next greater element** for every element. The next greater element of x is the first element to the right of x that is strictly greater than x. If there is no such element, write **-1** for it. Put the answers **back in the original array**.

Use a **stack implemented as a singly linked list** that stores integers.

Input:	`4 5 2 25`
Output:	`5 25 25 -1`
Input:	`13 7 6 12`
Output:	`-1 12 12 -1`

a) [4 Marks] Implement the stack. Make a class Node and a class Stack with push, pop, top, isEmpty and a destructor that frees the remaining nodes.

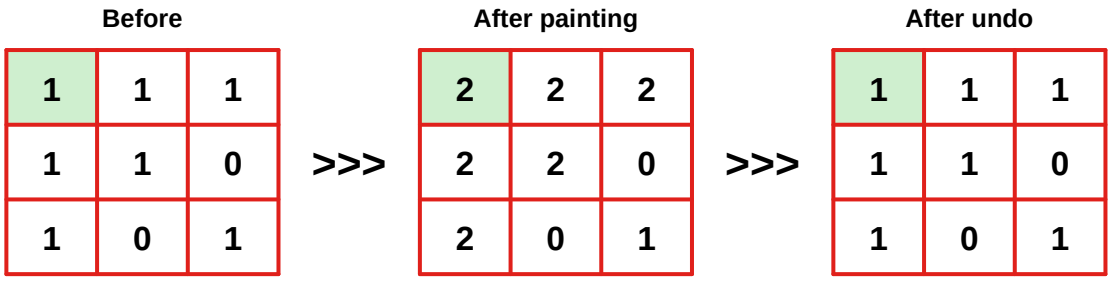
b) [6 Marks] Use your stack to find the next greater element of every element and update the original array with the answers.

```
void nextGreater(int arr[], int n)
```

CLO #: 2

Q2: [15 Marks]

An image is stored as an M x N matrix and every cell holds a color number. Starting from a given cell (sr, sc), paint that cell and every cell connected to it by moving up, down, left or right, that has the same color as the start cell, with **new_color**. The painting must be done using **recursion**. Example: a 3x3 matrix, start cell (0,0) and new_color = 2.



Start cell (0,0) is the green cell. new_color = 2. Cells with color 0 or cut off from the start cell stay the same.

You also have to be able to **undo** the painting. Use a **stack implemented as a singly linked list**, but this time every node stores more than one value: **oldColor, row** and **col**. Before a cell is painted, push its original state on the stack. Later the undo function uses the same stack to bring every changed cell back to its old color.

a) [3 Marks] Implement the stack. Make a class Node (oldColor, row, col) and a class Stack with push, pop, top, isEmpty and a destructor that frees the remaining nodes.

```
void push(int oldColor, int row, int col)
```

b) [6 Marks] Recursive function that paints the cells. Before updating a cell it must push the original state of that cell on the stack. If the start cell already has new_color, nothing should be painted.

```
void paintCells(int image[][MAX], int rows, int cols, int r, int c, int oldColor, int newColor, Stack& st)
```

c) [4 Marks] Undo function that pops the stack until it is empty and puts every cell back to its old color.

```
void undoPaint(int image[][MAX], Stack& st)
```

d) [2 Marks] In main, print the matrix before painting, after painting and after the undo.

Rules: Only the iostream library is allowed. No vectors, hashmaps or any other STL container. Use classes instead of structs. Write your own stack as a singly linked list, do not use the STL one.